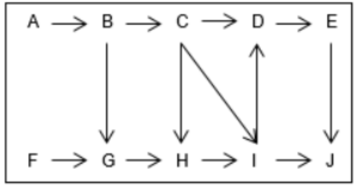
DIRECTIONS *for questions 1 to 5:* Answer these questions on the basis of the information given below.
The manufacturing of an electronic component comprises eight processes, P1 through P8, which should be carried out in a specific order. A factory has exactly eight machines, M1 through M8, which carry out the eight processes, P1 through P8, respectively.

These machines are arranged on a factory floor, the layout of which is given in the figure below. There are ten possible locations, A to J, on the factory floor and the eight machines are placed in eight of these ten locations. During the manufacturing of the electronic component, it has to be moved from one machine to another, based on the order in which the processes must be performed. The direction(s) in which a component can move between any two locations is indicated in the figure by the arrows. Further, it is known that the manufacturing process starts from one of the corner locations i.e. A, F, E, or J.



It is also known that

i. P1 must be started only after P2 ends.

ii. immediately after P7 ends, P5 must be started.

iii. each of P6 and P8 must be started only after P1 ends, and both P6 and P8 must end for P3 to start, which is not the final process.

iv. M3 is not in location D.

v. P4 must start only after P1 ends but must start before P7 begins.

Q1. DIRECTIONS *for questions 1 to 5:* Select the correct alternative from the given choices.
Which of the following locations need not have any machines?

- a) **A**
- b) **H**
- c) G
- d) E

Q2. DIRECTIONS *for questions 1 to 5:* Select the correct alternative from the given choices.
In how many ways could the machines have been placed on the factory floor?

- a) **2**
- b) **4**
- c) 6
- d) 8

Q3. **DIRECTIONS** *for questions 1 to 5:* Select the correct alternative from the given choices.
Which is the fifth process for manufacturing the component?

- a) **P6**
- b) **P4**
- c) P3
- d) P5

Q4. **DIRECTIONS** *for questions 1 to 5:* Select the correct alternative from the given choices.
Which of the following processes would occur immediately after P3?

- a) **P7**
- b) **P5**
- c) P1
- d) P4

Q5. DIRECTIONS *for questions 1 to 5:* Select the correct alternative from the given choices.
If location G has neither M6 nor M8, what BEST can be said about the location of M6?

- a) **It is in location C or location H.**
- b) **It is in location C or location I.**
- c) It is in location H.
- d) It is in location H or location I.

DIRECTIONS *for questions 6 to 10:* Answer these questions on the basis of the information given below.

A college admission test was attempted by 100 students during a particular year. The number of marks scored by the 100 students in the exam was a distinct even integer between 0 and 300 (both inclusive). All the 100 students were given ranks in the descending order of their ranks from 1 to 100.

However, a few days after providing these ranks to the students, it became known that there was an issue in one of the questions. After taking this issue into consideration, the marks of the 100 students were recalculated and based on the recalculated marks, revised ranks were provided to the 100 students. For any student, the marks of the student changed by 0 or 2 or 4 marks (increase or decrease). Further, the recalculated marks of the 100 students were also distinct even integers between 0 and 300 (both inclusive).

The following table provides the initial ranks and the revised ranks of six students, A through F, who were among the 100 students:

Student	Initial Rank	Revised Rank
A	70	73
B	1	3
C	29	31
D	2	1
E	30	32
F	27	24

It is also known that

i. the difference between the recalculated marks of A and C is 148.

ii. the difference between the recalculated marks of B and E is 94.

Q6. DIRECTIONS *for questions 6 to 8:* Select the correct alternative from the given choices.
Which of the following BEST describes the number of marks score by A before the recalculation?

- a) **Exactly 60**
- b) **Either 58 or 60**
- c) Exactly 58
- d) Either 56 or 58

Q7. DIRECTIONS *for questions 6 to 8:* Select the correct alternative from the given choices.
How many of the following can be the initial marks (i.e., prior to the recalculation) of the student whose revised rank is 20?

I. 230

II. 226

III. 260

IV. 266

V. 220

- a) **4**
- b) **3**
- c) 2
- d) 5

Q8. DIRECTIONS *for questions 6 to 8:* Select the correct alternative from the given choices.
Which of the following BEST describes the revised rank of the student whose initial rank is 31?

- a) **Either 29 or 30**
- b) **Exactly 30**
- c) Either 28 or 30
- d) Exactly 29

Q9. DIRECTIONS *for questions 9 and 10:* Type in your answer in the input box provided below the question.
Before the recalculation, the marks of the students whose initial ranks are from 33 to 50 formed an Arithmetic Progression.

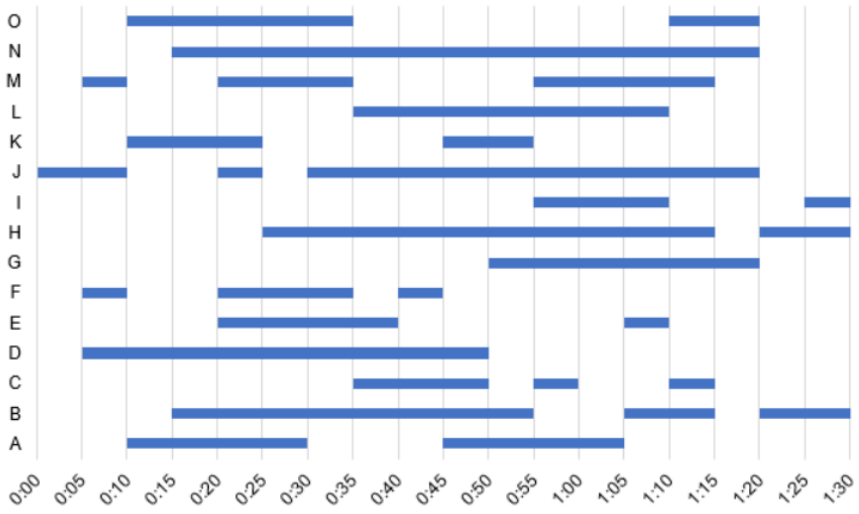
What is the maximum possible difference between the initial marks of the student whose initial rank is 40 and the student whose initial rank is 42?

Q10. DIRECTIONS *for questions 9 and 10:* Type in your answer in the input box provided below the question.
Before the recalculation, the marks of the students whose initial ranks are from 33 to 50 formed an Arithmetic Progression.

If the common difference of the Arithmetic Progression is the maximum possible, what is the maximum possible initial marks of the student whose initial rank is 55?

DIRECTIONS *for questions 11 to 15:* Answer these questions on the basis of the information given below.

Sachin watched a movie which was exactly 90 minutes long. There were exactly fifteen actors, A through O, in the movie and these actors appeared on screen at different times during the movie. Sachin tracked the actors who appeared on screen at different points of time during the runtime of the movie. The following graph provides the details of when each actor appeared on screen during the runtime of the movie, with the horizontal axis representing the runtime (in hh:mm) and the vertical axis representing the actors in the movie:



Q11. DIRECTIONS *for question 11:* Select the correct alternative from the given choices.

For how many minutes in the movie were there exactly five actors on screen?

a) **0** ✓ Your answer is correct

b) **5**

c) 10

d) 15

Q12. DIRECTIONS *for questions 12 and 13:* Type in your answer in the input box provided below the question.

What is the maximum number of actors who appeared on screen at any given point of time in the movie?

Q13. DIRECTIONS *for questions 12 and 13:* Type in your answer in the input box provided below the question.

If there are more than eight actors on screen at the same time, then the movie is said to have been crowded for that duration. For how long (in minutes) was the movie crowded?

Q14. **DIRECTIONS** *for question 14:* Select the correct alternative from the given choices.
The names of the actors were displayed at the end of the movie in the same order in which they first appeared on screen in the movie. If two or more actors first appeared on screen at the same time, the names of these actors were displayed in the descending order of the durations for which the actors appeared in the movie.

What is the ninth name to be displayed in the movie?

a) **N**

b) **B** ✓**Your answer is correct**

c) E

d) O

Q15. **DIRECTIONS** *for question 15:* Type in your answer in the input box provided below the question.

The names of the actors were displayed at the end of the movie in the same order in which they first appeared on screen in the movie. If two or more actors first appeared on screen at the same time, the names of these persons were displayed in the descending order of the total duration for which each person appeared in the movie.

Considering only the actors whose names were the last nine to be displayed at the end of the movie, what is the maximum duration (in minutes) for which any actor appeared on screen in the movie?

DIRECTIONS *for questions 16 to 20:* Answer these questions on the basis of the information given below.

Eight persons – Anurag, Bhim, Chintu, Dev, Pranav, Rajiv, Satish and Varun – live on different floors in a building which has eight floors from first floor to eighth floor. The building also has a basement which is one floor below the first floor of the building. The building has exactly one lift which connects all the eight floors and the basement. On each day, each person leaves for office either at 8:00 AM or 9:00 AM or 10:00 AM. All the persons who leave for office at a particular time enter the lift at their respective floors and reach the basement together. Each of the eight persons, when leaving for office, board the lift only when it is on its way down. Any two persons are said to share the lift for one floor if they are in the lift together for one floor. For example, if two persons are in the lift when the lift went from first floor to the basement, then the two persons are said to have shared the lift for one floor.

Further, it is also known that,

i. not more than three people leave for office at the same time and no two persons share the lift for more than five floors.

ii. Anurag does not leave for office at 9:00 AM and he lives on the floor immediately above the one on which Dev lives.

iii. Varun does not leave for office at 10:00 AM.

iv. the person who lives on the floor three floors below the one on which Anurag lives leaves for office at 9:00 AM.

v. the person who lives on the third-floor leaves for office at 10:00 AM and neither is this person Satish nor does he share the lift with Satish.

vi. the person who lives on the fourth-floor leaves for office at 9:00 AM and he shares the lift with Pranav, who lives on the second floor.

vii. Chintu, who lives on one of the floors above the one on which Anurag lives, shares the lift with Bhim, while Dev shares the lift for five floors with a person who leaves at 8:00 AM.

Q16. DIRECTIONS *for questions 16 to 20:* Select the correct alternative from the given choices.
Who lives on the fourth-floor of the building?

- a) **Dev**
- b) **Varun**
- c) Bhim
- d) Either Bhim or Rajiv

Q17. **DIRECTIONS** *for questions 16 to 20:* Select the correct alternative from the given choices.
Who among the following shares the lift with the least number of persons?

- a) **Dev**
- b) **Bhim**
- c) Rajiv
- d) Satish

Q18. DIRECTIONS *for questions 16 to 20:* Select the correct alternative from the given choices.
At what time does the person who lives on the eighth-floor leave for office?

- a) **8:00 AM**
- b) **9:00 AM**
- c) 10:00 AM
- d) Either 9:00 AM or 10:00 AM

Q19. DIRECTIONS *for questions 16 to 20:* Select the correct alternative from the given choices.
Who among the following lives on a floor above the one on which Rajiv lives and leaves for office at the same time as him?

- a) **Bhim**
- b) **Anurag**
- c) Satish
- d) Either Anurag or Satish

Q20. DIRECTIONS *for questions 16 to 20:* Select the correct alternative from the given choices.
If Satish leaves for office one hour before Chintu leaves for office, who lives on the first-floor?

a) **Satish**

b) **Rajiv**

c) Either Rajiv or Varun

d) Either Satish or Varun